

TraceMind — investor brief

For: a friend who's looking. Be honest, not polished. **Date:** 2026-05-06 **Author:** Aaditya **Status:** Pre-seed, no raise yet, asking for thinking time + intros, not a check.

Slide 1 — One line

TraceMind is a local-only memory OS for humans and AI agents. Captures intent, predicts outcomes, and earns trust on-device. Cloud memory is the default; we're betting privacy is the moat.

Slide 2 — The problem (in one paragraph)

Every AI assistant today forgets. The fixes on the market — Mem0, Honcho, Letta, Zep, SuperLocalMemory — solve forgetting by sending your data to their cloud. That's a fine MVP. It's a terrible long-term position once "memory" becomes the most sensitive surface in your digital life: meeting notes, intents, decisions, side comments, clipboard. The data moat goes the wrong direction.

The opening: the user is already on a powerful machine. The right answer is to build the brain there, and only there.

Slide 3 — Our wedge

We are not a "memory database." We are a **system of intents**.

The primitive is a **Commitment** — a recorded statement of intent or decision with metadata you can predict over later:

```
kind          intent | decision | hypothesis
stakes        low | medium | high | reversible
horizon       when this should resolve by
expected      what success looks like
```

You commit to something. Time passes. You resolve it. The polarity becomes training data for a personal world model.

Why this is the wedge: notes apps are a saturated category. Decision-tracking with a model that learns *your* track record is not. And it composes naturally with any LLM — we expose 21 tools over MCP.

Slide 4 — How it works (the brain in 4 layers)

```
Sensory      → clipboard / shell / MCP intake
buffer       (tm-capture daemon, always-on)
```

Memory	→ 6 layers (working / episodic / semantic / procedural / sensory / intent) – all on-device SQLite
Cognition	→ 10 ops (attention / recall / association / reasoning / consolidation / dreaming / pattern detection / synthesis / self-monitor / world modelling)
Surfaces	→ CLI · MCP server (21 tools) · capture daemon · Tauri desktop app

- **17 Rust crates**, ~12k LOC, 163 tests passing.
- **All data in `~/.tracemind/`** — one SQLite file plus a JSONL audit trail. The user can `rm -rf` it. That's the contract.
- **No network in the request path.** Network is opt-in, only for initial model downloads.

Slide 5 — What's shipped

Memory pipeline (Act 1 of the demo):

- BGE-small ONNX embeddings (384-dim, real, on-device)
- GLiNER NER for entity extraction
- SQLite graph + vector store
- Immutable trace log (every ingest gets a UUID)
- ColBERT reranking auto-downloads on first use

Intent system (Act 2):

- **Commitment + Outcome + Anticipation** types
- CLI: `tracemind commit / commitments / resolve / brief`
- MCP tools: `memory_commit`, `memory_resolve`, `memory_brief`
- Daily brief surface — calm-mode, no nagging

World model (Act 3-4):

- Logistic regression on 22-dim metadata feature space
- Auto-retrains on every resolve
- Calibration view that goes silent when miscalibrated
- (This is the "earned trust" loop)

Integration (Act 5):

- MCP server with 21 JSON-RPC tools
- Stdio-only — no network, no daemons
- Any MCP-aware client (Claude Code, Cursor, custom agents) plugs in

The proof: there's a 6-minute screen recording of all five acts running on real data with real embeddings. I'll send it.

Slide 6 — What's good (honest version)

1. **The architecture is right.** 6 memory types and 10 cognitive ops match the brain literature, not a database schema. Hard to copy without rebuilding from scratch.
 2. **Local-only is real, not marketed.** Every storage layer is SQLite on the user's machine. No "encrypted cloud sync that's actually a default-on cloud sync."
 3. **Real eval harness.** We run LoCoMo (the published memory benchmark) with a CI gate that fails any PR dropping >0.5 F1. That's discipline most pre-seed AI teams don't have.
 4. **The wedge is sharp.** "System of intents" is a category that doesn't exist yet. Notes apps are saturated; commitment-tracking with a personal predictive model is empty.
 5. **The brain extends naturally.** Same crates power the CLI, the MCP server, the capture daemon, and the desktop app. No architectural rewrite when we add a new surface.
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Slide 7 — What's bad (also honest)

1. **LoCoMo F1 is 25.7. The target is 85.** That's a real gap. Mem0 / Letta / Zep clear 70+ because they use cloud LLMs as their synthesis layer. We've been on Tier-0 (extractive, template-based) the whole time. The fix is shipping Tier-1 (Qwen 2.5 1.5B Q4, ~900MB on-device LLM, already scaffolded). This is the single biggest lever.
 2. **The world model is logistic regression.** It works (calibration loop runs, brief shows the outlook line), but logistic regression on 22 features is not "an AI moat." We have an MLP + held-out validation in WIP, but it's not landed yet.
 3. **GLiNER NER had a non-determinism paper cut.** (*fixed 2026-05-06*) Same sentence, different runs, used to give 0 to 5 entities. Root cause: ORT thread scheduling + a 0.30 sigmoid cutoff sitting right where logits flutter. Fix landed: deterministic ORT session (single-thread, deterministic kernels), 0.05 threshold deadband, stable NMS tiebreakers. Costs ~1 F1; buys reproducible output.
 4. **No multi-user, no sync, no collaboration.** By design today, but every investor will ask "team memory?" We need a clean answer: Phase 6, ACL + REST API.
 5. **Tauri desktop app builds but doesn't ship.** Pre-existing build script issue. We exclude it from CI test runs. Not investor-blocking, but not polished.
 6. **No funded user base.** GitHub stars are not customers. The honest read is we're optimizing the engine before we have proof that anyone wants this engine. (Counter-read: privacy-first is a "you don't realize you want it until cloud memory burns you" category — we're early on purpose.)
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Slide 8 — What needs to improve (the cognition roadmap)

Ten moves, ranked by impact / effort. Full version in docs/PHASE4_DELIGHT.md §9.

Tier 1 — biggest unlocks (do first)

- 1. **Ship Tier-1 synthesis (Qwen 2.5 1.5B Q4).** Single biggest lever. Every cognition op looks 3× smarter for free. Effort: small.
- 2. **Real reward signal into the bandit.** Today the bandit is starving on weak proxies. One keypress feedback in the brief compounds for weeks. Effort: small.
- 3. **Query rewriting before recall.** Tier-1 expands "how does X earn trust?" into 3 queries, fuses with RRA. ~5–8 F1 on multi-hop. Effort: small.

Tier 2 — structural fixes

- 4. Surprisal-gated ingestion (skip low-info captures).
- 5. Reasoning chains with beam search, not greedy.
- 6. Calibration via Platt scaling on world-model probabilities.

Tier 3 — moat work

- 7. Replace WL-kernel association with a learned graph embedding (GNN).
- 8. Consolidation via Louvain/Leiden community detection nightly.
- 9. Pattern detector → factorization machine over interaction features.
- 10. Working-memory ring (last N turns + retrievals fed back).

The one to bet on: #1, Tier-1 synthesis. Everything else is a multiplier on a base. Without Tier-1 the base is templates.

Slide 9 — The plan (sprints A–G, canonical in INTENT_SYSTEM.md §10)

Sprint	Focus	Status
A	Intent primitives + MCP tools	DONE
B	Working memory ring + capture miner	IN FLIGHT
C	World model (logistic) + calibration loop	DONE
D	Pattern detector + L2/L3 anticipations	NEXT
E	Tier-1 synthesis default-on (THE big sprint)	HIGHEST ROI
F	Dreaming / consolidation / surprisal	DESIGNED
G	Companion UX (voice, narrative, garden)	SCOPED

The cognition roadmap items map onto these — see PHASE4_DELIGHT.md §9.6 for the mapping table.

Slide 10 — The moat (in three lines)

- 1. **Privacy by construction.** Not a feature, not a setting. The data physically lives on the user's machine.

2. **Brain-shaped architecture.** 6 memory types × 10 cognitive ops is hard to clone without a ground-up rewrite. Cloud memory competitors built database schemas; we built an architecture.
 3. **The trust loop.** The world model goes silent when it's miscalibrated. That's the opposite of every notification-driven product. Earning the right to speak, by performance, is the long-term differentiator.
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Appendix — proof points

- **Code:** 17-crate Rust workspace, 163 tests passing on [main](#).
- **Eval:** LoCoMo mini F1 25.70 (BGE), CI-gated.
- **Docs:** four canonical, all current
 - [docs/PHASE4_DELIGHT.md](#) — current progress + cognition roadmap
 - [docs/INTENT_SYSTEM.md](#) — wedge spec + 7-sprint plan
 - [docs/BRAIN_ARCHITECTURE.md](#) — full architecture
 - [docs/LOCOMO_RESULTS.md](#) — eval baselines
- **Demo:** 6-minute screen recording in [demo/recordings/walkthrough-clean-20260506-215825.mov](#)
- **Repo:** [github.com/aadi2vec/tracemind](#) (private; ask for access)